Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION 2025

Name of the Course: **B. Tech** — Semester: **2nd**

Name of the Paper: **Advanced Organic Chemistry** — Paper Code: **TCH202**

Time: 3 Hours — Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the techniques of simple distillation and distillation under reduced pressure, providing appropriate diagrams where needed.	**CO1**
(b)	Explain the principles and procedures of steam distillation and solvent extraction, with suitable diagrams.	
(c)	Illustrate the crystallization and sublimation method with suitable example	
Q2	(20 marks)	
(a)	Explain the effect of substituents on electrophilic substitution reactions with appropriate examples.	**CO2**
(b)	Define the SN^1 reaction and describe its mechanism. Which type of alkyl halide undergoes SN^1 reaction more readily, and why?	
(c)	Describe the nitration and halogenation of benzene with detailed reaction mechanisms.	
Q3	(20 marks)	
(a)	Provide a brief overview of nanoparticles in terms of their types and applications	**CO3**
(b)	Describe different methods for synthesizing nanoparticles and discuss the health risks associated with toxic chemicals used in these processes.	
(c)	List and briefly explain the twelve principles of green chemistry.	
Q4	(20 marks)	
(a)	Describe the principle, advantages, and working procedure of Thin Layer Chromatography (TLC).	**CO4**
(b)	Discuss Column Chromatography with reference to its principle, working methodology, and applications.	
(c)	Define volumetric analysis and describe the various types of titrations with appropriate examples.	
Q5	(20 marks)	
(a)	Give an overview of carbohydrates and explain their classification with suitable examples.	**CO5**
(b)	Describe different methods for preparing glucose and explain its main physical and chemical properties	
(c)	Explain the Killiani–Fischer synthesis and Ruff degradation with suitable examples.	